



$$L_x = I \omega_x$$

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

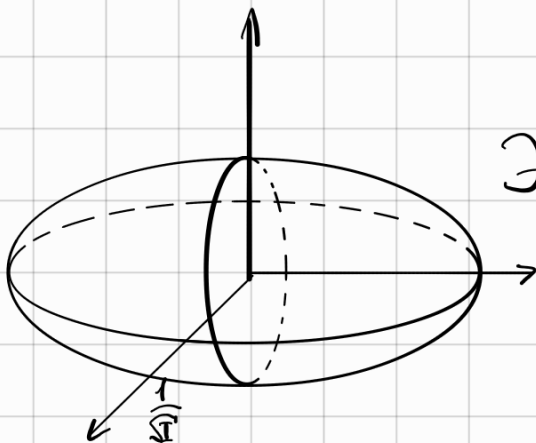
↑  
ТЕНЗОР ИНЕРЦИИ

$$I_{xx} = \int (y^2 + z^2) dm$$

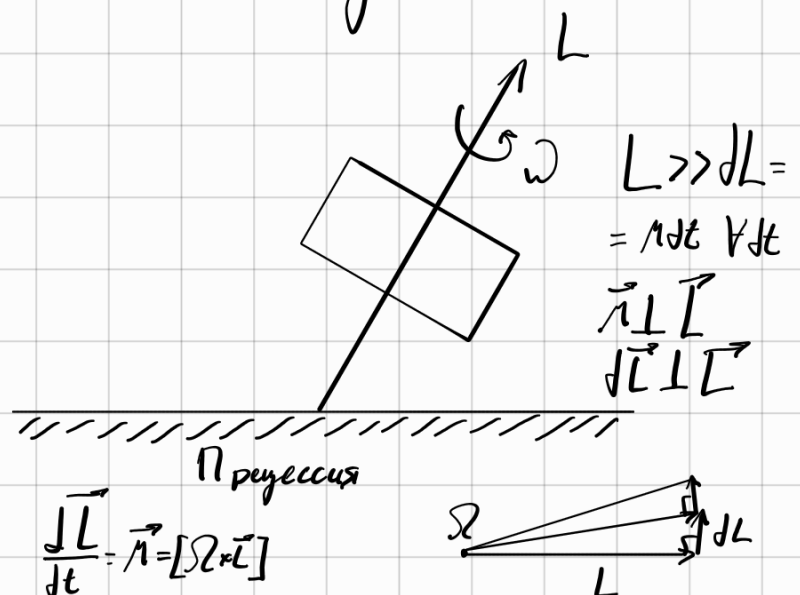
$$I_{xy} = - \int xy dm = I_{yx}$$

ТЕНЗОР - симм. матр.  $\Rightarrow$   $\exists$  главные оси:

$$\begin{pmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{pmatrix}$$



Эллипсоид инерции



$$L \gg dL = \frac{dL}{dt} dt$$

$$\vec{L} \perp \vec{\omega}$$

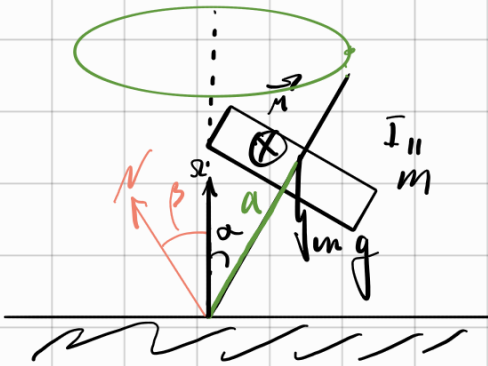
$$\frac{d\vec{L}}{dt} = \vec{\Omega} \times \vec{L}$$



11.1

$$N_y = mg$$

$$N = mg a \sin \alpha$$



$$N_x = m \Omega^2 a \sin \alpha$$

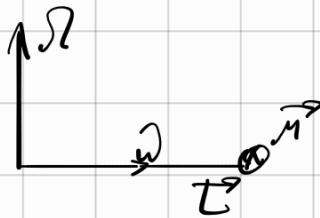
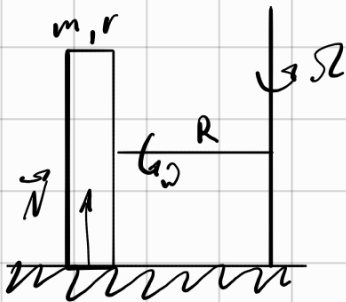
$$\tan \beta = \frac{N_x}{N_y} = \frac{\Omega^2 a \sin \alpha}{g}$$

$$L = I_{\parallel} \Omega$$

$$M = \Omega \cdot I_{\parallel} \Omega \cdot \sin \alpha$$

$$\Omega = \frac{mg a}{I_{\parallel}}$$

11.14



$$M = FR$$

$$\Omega \cdot L_c = \Omega \frac{m R^2}{2} \Omega$$

$$\Omega = \Omega \frac{R}{r}$$

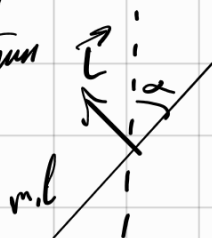
$$F = \frac{m \Omega^2 r}{2}$$

$$N = mg + \frac{m \Omega^2 r}{2}$$

11.24

$$n = 3000 \frac{1}{\text{mm}}$$

$$\alpha = 89.9^\circ$$



$$dm = \frac{m}{l} dx$$

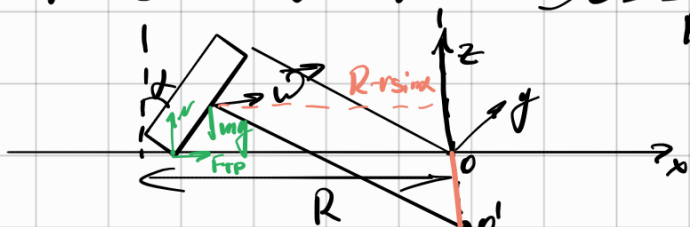
$$dL = x dp = x dm \Omega \sin \alpha$$

$$L = \int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{m}{l} \Omega \sin \alpha x^2 dx = \frac{m}{l} \Omega \sin \alpha \frac{x^3}{3} \Big|_{-\frac{l}{2}}^{\frac{l}{2}} =$$

$$= \frac{m l^2}{12} \Omega^2 \sin \alpha$$

$$M = \Omega L \cdot \sin\left(\frac{\pi}{2} - \alpha\right) = \frac{m l^2}{12} \Omega^2 \sin \alpha \sin\left(\frac{\pi}{2} - \alpha\right)$$

11. 18  $\omega = \frac{v}{r}$   $\Omega = \frac{v}{R}$



$$\vec{\tau} = [\vec{L} \times \vec{L}] = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & -\Omega \\ -L_y & 0 & L_z \end{vmatrix} = \vec{j} \Omega L_x$$

$$m \Omega^2 (R - r \sin \alpha) = F_{TP}$$

$$N = mg$$

$$M_N = mg R$$

$$M_{mg} = -mg (R - r \sin \alpha)$$

$$M_{TP} = m \Omega^2 (R - r \sin \alpha) r_{od} = m \Omega^2 (R - r \sin \alpha) \frac{R \sin \alpha - r}{\cos \alpha}$$

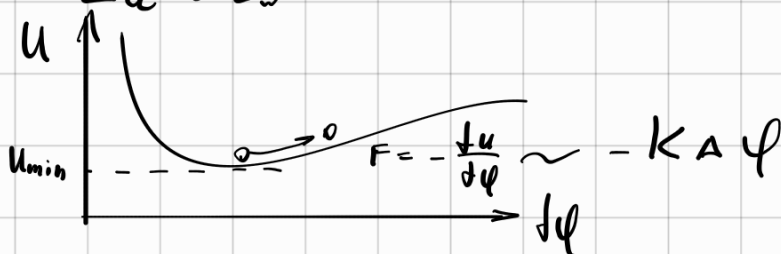
$$\Omega L_x = - \frac{(R - r \sin \alpha)(R \sin \alpha - r)}{\cos \alpha} m \Omega^2 - (R - r \sin \alpha) mg + mg R$$

$$\vec{L} = \vec{L}_{||} + \vec{L}_{\perp}$$

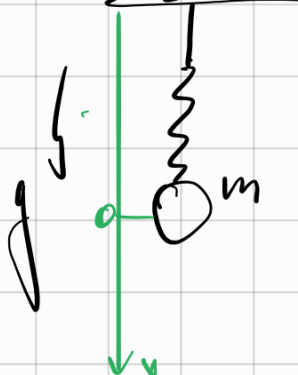
$$L_{||} = I_{||} \Omega_{||} = m r^2 (\omega + \Omega_{||})$$

$$L_{\perp} = I_{\perp} \Omega_{\perp} = \left( \frac{m r^2}{2} + m \left( \frac{(R - r \sin \alpha)^2}{\cos \alpha} \right) \right) \Omega_{\perp}$$

$$L_{\Omega} \sim L_{\omega}$$



543



$$F = K(\ell - \ell_0)^2$$

$$mg = K \Delta \ell_0^2$$

$$\Delta \ell_0 = \sqrt{\frac{mg}{K}}$$

$$F_y = K(\Delta \ell_0 + x)^2$$

$$m \ddot{x} = mg - K(\Delta \ell_0 + x)^2$$

$$m\ddot{x} = -2K\Delta l_0 x$$

$$\ddot{x} + \frac{2K\Delta l_0}{m} x = 0$$

$$\omega = \sqrt{\frac{2K\Delta l_0}{m}} = \sqrt{\frac{4K\ell}{m}}$$

$$U = \frac{A\dot{\varphi}^2}{2}$$

$$K = B \frac{\dot{\varphi}^2}{2}$$

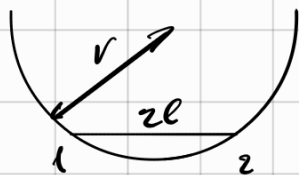
$$E = A \frac{\dot{\varphi}^2}{2} + B \frac{\dot{\varphi}^2}{2}$$

$$E = 0 = A\dot{\varphi}\ddot{\varphi} + B\dot{\varphi}\ddot{\varphi}$$

$$0 = A\ddot{\varphi} + B\ddot{\varphi}$$

$$\ddot{\varphi} + \frac{A}{B}\varphi = 0 \quad \omega = \sqrt{\frac{A}{B}} \quad T = 2\pi\sqrt{\frac{B}{A}}$$

5.22



1)  $\perp \rho_{uc}$



$$\rho = \sqrt{r^2 - \ell^2}$$

$$U = 2mg(\rho - \rho \cos \varphi) \approx 2mg\rho \frac{\varphi^2}{2}$$

$$K = \frac{2m\rho^2}{2} \dot{\varphi}^2$$

$$T = 2\pi\sqrt{\frac{B}{A}} = 2\pi\sqrt{\frac{2m\rho^2}{2mg\rho}} = 2\pi\sqrt{\frac{r^2 - \ell^2}{g}}$$

2)



$$U = 2mg\rho \frac{\varphi^2}{2}$$

$$K = 2m r^2 \frac{\dot{\varphi}^2}{2}$$

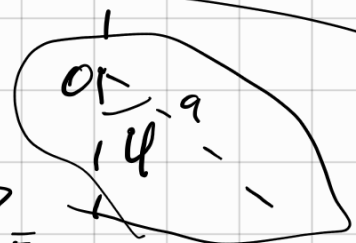
$$T = 2\pi\sqrt{\frac{K}{U}}$$

Задача 5.23. МАСТНИК

$$I\ddot{\varphi} = -mga \sin \varphi$$

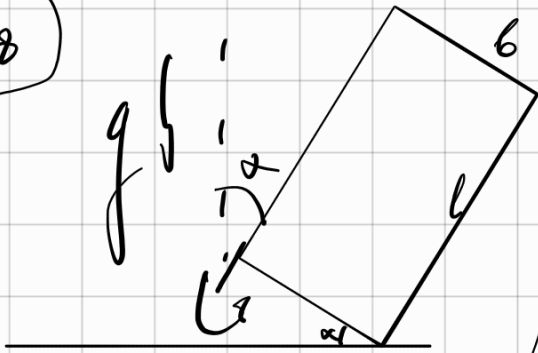
$$\ddot{\varphi} + \frac{mga}{I}\varphi = 0$$

$$T = 2\pi\sqrt{\frac{I}{mga}}$$



$$= 2\pi \sqrt{\frac{L_{OP}}{g}} \quad L_{OP} = \frac{I}{ma}$$

1048



T-?

$$\frac{b}{2} (1 - \cos \varphi)$$

$$\frac{b}{2} \sin \alpha (1 - \cos \varphi)$$

$$U \approx mg \frac{b}{2} \sin \alpha \frac{\varphi^2}{2}$$

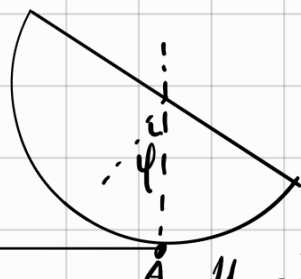
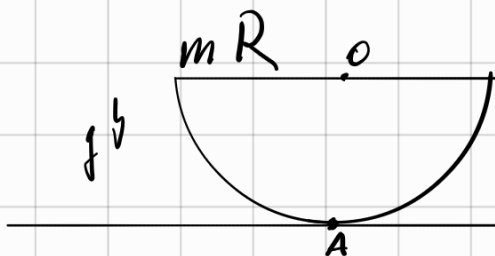
$$K = I \frac{\dot{\varphi}^2}{2}; \quad I = \frac{mb^3}{3}$$

$$T = 2\pi \sqrt{\frac{2b}{3g \sin \alpha}}$$

$$E = \frac{A\dot{\varphi}}{2} + \frac{B\dot{\psi}}{2}$$

$$T = 2\pi \sqrt{\frac{B}{A}}$$

10.84



$$K = I_A \frac{\dot{\varphi}^2}{2} \quad I_A = I_C + m(R - y_c)^2 + (y_c)^2$$

$$I_O = I_C + m y_c^2 = \frac{m R^2}{2}$$

$$I_A = \frac{m R^2}{2} - m y_c^2 + m(R - y_c)^2$$

$$= \frac{3}{2} m R^2 - \frac{8}{3\pi} m R^2$$

$$T = 2\pi \sqrt{\frac{m R^2 (\frac{3}{2} - \frac{8}{3\pi})}{\frac{4}{3\pi} m g R}} = 2\pi \sqrt{\frac{R}{g} \frac{3\pi - 16}{8}}$$

$$U = m g y_c (1 - \cos \varphi) = m g y_c \frac{\varphi^2}{2}$$

$$y_c = \frac{1}{m} \int_0^R y dm \quad dm = \frac{2 \sqrt{R^2 - y^2} m dy}{\pi R^2}$$

$$y_c = \frac{2}{\pi R^2} \int_0^R y \sqrt{R^2 - y^2} dy = \frac{-2}{\pi R^2} \int_0^R x dx$$

$$R^2 - y^2 = x$$

$$dx = -2y dy$$

$$= -\frac{2}{\pi R^2} \frac{2}{3} x^{\frac{3}{2}} \Big|_0^R = \frac{4}{3\pi} R$$

$$\vec{a} = \frac{\vec{F}}{m} \quad \text{в УГО}$$

$$m\vec{a} = \vec{F} + \vec{F}_{\text{упр}} \quad \vec{F}_{\text{упр}} = -m\vec{a}_{\text{УГО}}$$

0 12.1

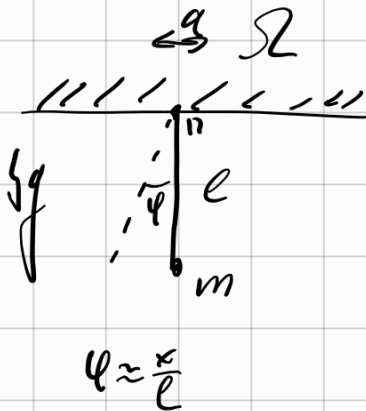
Дано:

$$l = 9,8 \text{ см}$$

$$\Omega = 11 \frac{\text{рад}}{\text{с}}$$

$$\alpha = 1 \text{ м}$$

Найти: A



$$x_n = a \cos \Omega t$$

$$F_{\text{упр}} = -m \ddot{x}_n = m a \Omega^2 \cos \Omega t$$

$$m \ddot{x} = m a \Omega^2 \cos \Omega t - T \cdot \frac{x}{l} \quad \ominus$$

$$\psi \text{ мал } \cos \psi \approx 1$$

$$\ominus m a \Omega^2 \cos \Omega t - m g \frac{x}{l}$$

$$\ddot{x} + g \frac{x}{l} = a \Omega^2 \cos \Omega t$$

неоднородное Д.У.

общее решение:  $\Sigma$  общее решение однородного

+ частное решение неоднородного  
в виде  $A \cos(\Omega t + \psi)$

(1)

$$-\Omega^2 A \cos(\Omega t + \psi) + \omega_0^2 A \cos(\Omega t + \psi) = a \Omega^2 \cos \Omega t$$

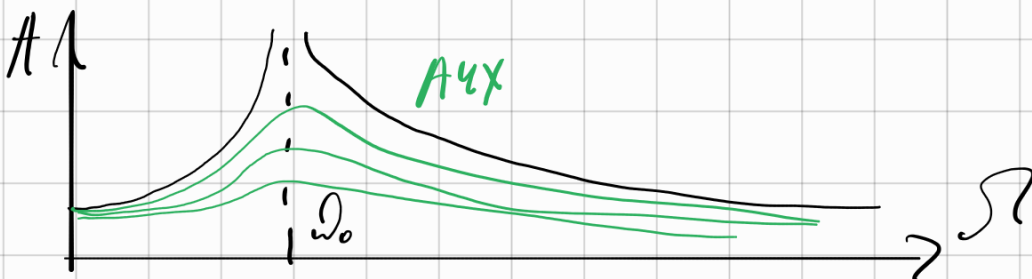
$$-\Omega^2 A \cos \Omega t \cos \psi + \Omega^2 A \sin \Omega t \sin \psi + \omega_0^2 A \cos \Omega t \cos \psi -$$

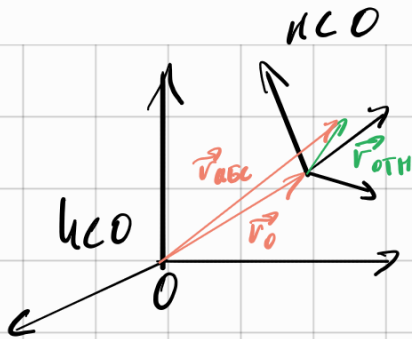
$$- \omega_0^2 A \sin \Omega t \sin \psi = a \Omega^2 \cos \Omega t$$

$$\begin{cases} -\Omega^2 A \cos \psi + \omega_0^2 A \cos \psi = \Omega^2 a \\ \Omega^2 A \sin \psi - \omega_0^2 A \sin \psi = 0 \end{cases}$$

$$\sin \psi = 0 \quad \cos \psi = \pm 1$$

$$A = \frac{a \Omega^2}{\pm (\omega_0^2 - \Omega^2)} \quad \begin{matrix} \cos \psi = -1 \\ A \geq 0 \end{matrix} = \frac{a}{1 - \frac{\Omega^2}{\omega_0^2}} = 5,8 \text{ мм}$$





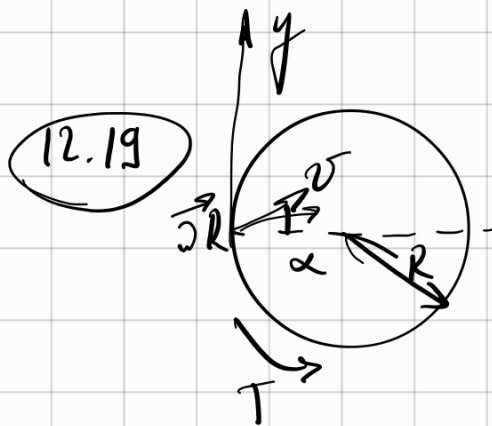
$$\vec{r}_{ABL} = \vec{r}_{OTH} + \vec{r}_0$$

$$\vec{v}_{ABL} = \vec{v}_{OTH} + [\vec{\omega} \times \vec{r}_{OTH}] + \vec{v}_0$$

$$\vec{a}_{ABL} = \vec{a}_{OTH} + \vec{a}_0 + [\vec{\omega} \times \vec{v}] + [\vec{\omega} \times [\vec{\omega} \times \vec{r}_{OTH}]] + 2[\vec{\omega} \times \vec{v}_{OTH}]$$

$$F_y \delta = m \omega^2 r_{OTH}$$

$$\vec{F}_{KOP} = 2m[\vec{v}_{OTH} \times \vec{\omega}] = -m\vec{a}_{KOP}$$



$$v \gg \omega R$$

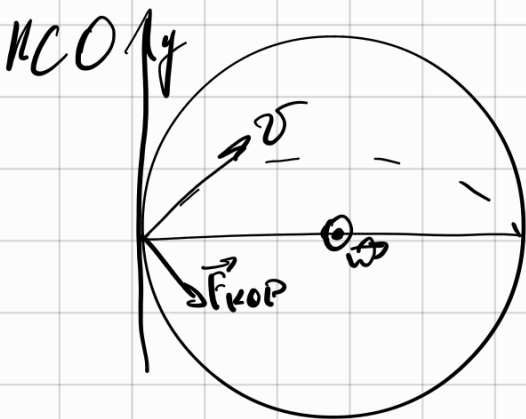
$$\alpha \ll 1$$

$$t \approx \frac{2R \cos \alpha}{|\vec{v} + \vec{\omega} R|} = \frac{2R(1 - \frac{\alpha^2}{2})}{v} \approx \frac{2R}{v}$$

$$|y(t)| = (v \sin \alpha - \omega R) t \approx \omega R t$$

$$\sin \alpha \approx \alpha = \frac{2\omega R}{v} \quad T = \frac{2\pi}{\omega}$$

$$\alpha = \frac{4\pi R}{vT} \approx 0.0209 \text{ rad} = 1.2^\circ$$



$$F_{KOP} = 2m\omega v$$

$$F_y \delta = m\omega^2 x \leq m\omega^2 R \ll m\omega v$$

$$m\ddot{y} = -2m\omega v$$

$$y(t) = v \sin \alpha - 2\omega v t$$

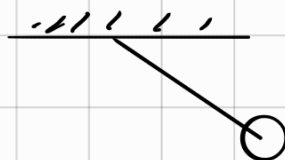
$$y(t) = v \sin \alpha t - 2\omega v t^2$$

$$t = \frac{2R}{v} \quad y(t) = 0$$

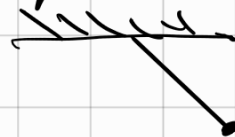
$$\alpha = \frac{\omega v}{v} \cdot \frac{2R}{v} = \frac{2\omega R}{v}$$

10.43

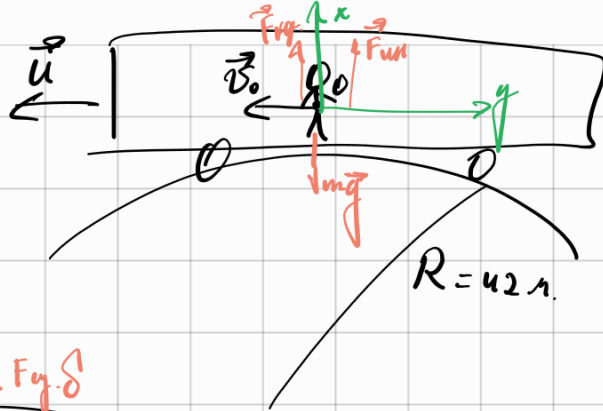
$$K_f = \frac{1}{2} \frac{v^2}{\omega^2} = \frac{1}{2} \frac{(\psi \epsilon)^2}{\omega^2}$$



Риск флуктуация попутного



12.27



Ocu bany bueme c gburu.  
abmodya  $\omega = \frac{u}{R}$

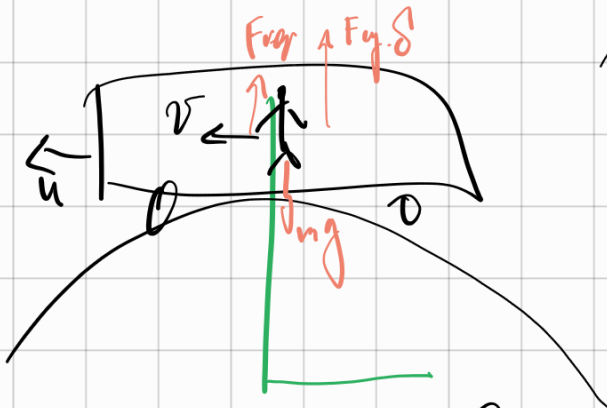
$$F_y \cdot S = 0$$

$$F_{\text{mag}} = 2m [v \times \omega]$$

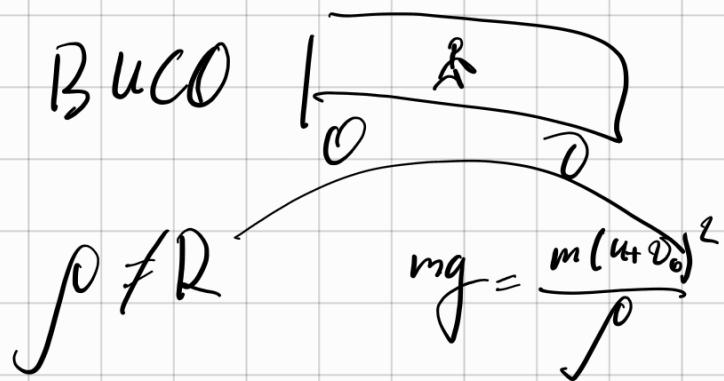
$$F_{\text{un}} = m \frac{u^2}{R}$$

$$mg = m \frac{u^2}{R} + 2m v_0 \cdot \frac{u}{R}$$

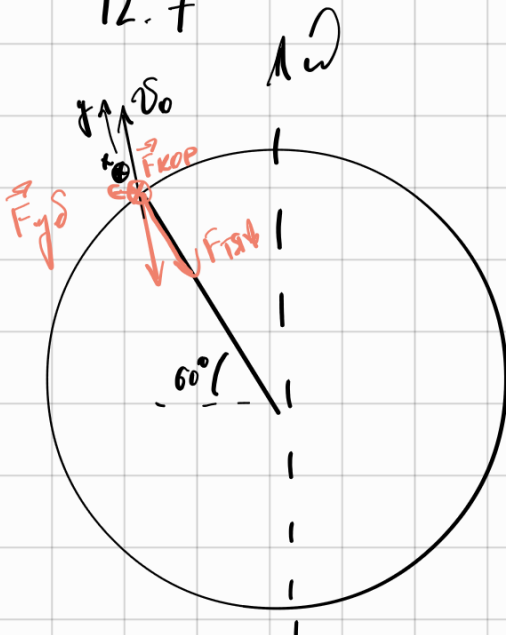
$$v_0 = \frac{gR - u^2}{2u} \approx 0.23 \text{ m/s}$$



$$mg = m \omega^2 R + 2m v_0 \omega$$



12.7

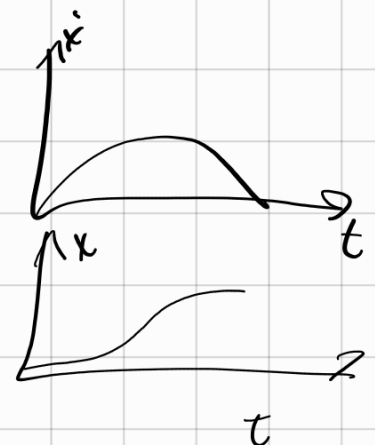


$$m \ddot{y} \approx mg$$

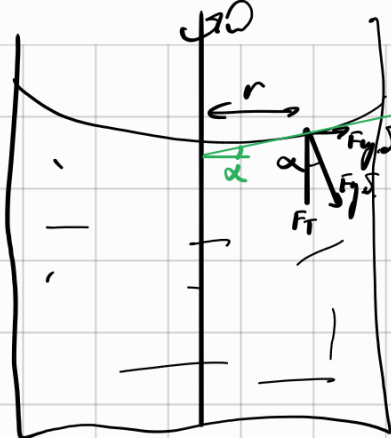
$$m \ddot{x} = 2m \dot{y} \omega \sin\left(\frac{\pi}{2} - \psi\right)$$

$$\dot{y} = v_0 - gt$$

gbanuon



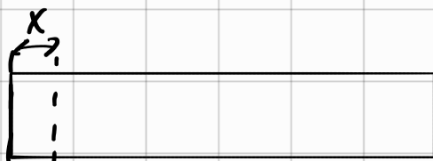
14.16



$$z(r) = ?$$

$$\tan \alpha = \frac{F_{ty} \delta r}{F_r} = \frac{\partial z}{\partial r}$$

$$z(r) = h + \frac{\partial^2 r^2}{2g}$$



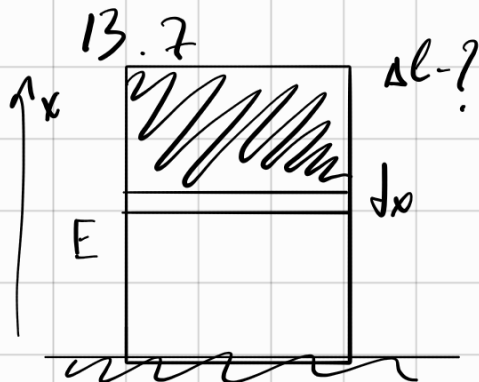
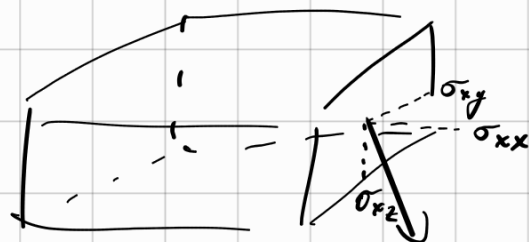
$$F = kx$$

$$\sigma = \frac{F}{S} \text{ механическое напряжение}$$

$$\epsilon = \frac{x}{l} \text{ относительная деформация}$$

$$\sigma = \epsilon \cdot E \text{ закон Гука}$$

$$\mu = -\frac{\epsilon_y}{\epsilon_x} \text{ коэфф. Пуассона}$$



$$P(x) = P(1 - \frac{x}{h})$$

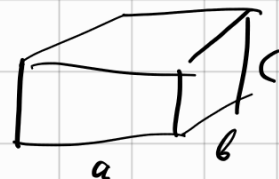
$$dE = \frac{d(\Delta l)}{dx} = \frac{P(x)}{SE}$$

$$\Delta l = \int_0^h \frac{P}{SE} (1 - \frac{x}{h}) dx = \frac{Ph}{2SE}$$

$$\frac{\Delta V}{V} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c} =$$

$$= \frac{\Delta a}{a} (1 - 2\mu)$$

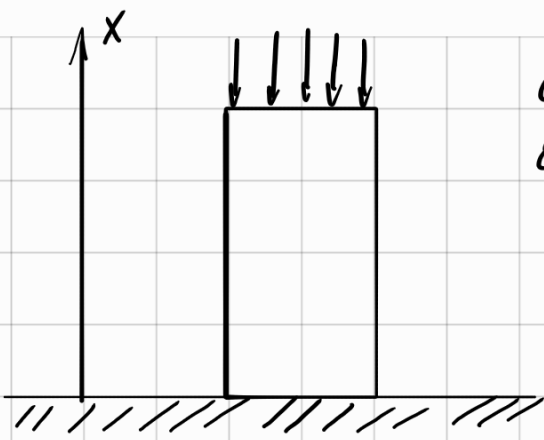
$$\frac{\Delta V}{V} = \frac{P}{2SE} (1 - 2\mu)$$



$$W = \frac{kx^2}{2}$$

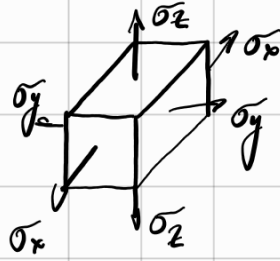
$$w = \frac{W}{V} = E \frac{\epsilon^2}{2} = \frac{\sigma^2}{2E} \text{ удельная потенциальная энергия}$$

$$W = \int_0^h w_{(x)} S dx = \frac{P^2 h}{6SE}$$



$$\epsilon_x = \frac{\sigma_x}{E}$$

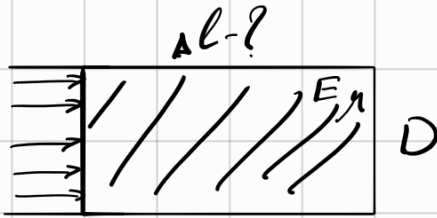
$$\epsilon_y = \epsilon_z = -\mu \epsilon_x = -\mu \frac{\sigma_x}{E}$$



$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E} = \frac{1}{E}(\sigma_x - \mu(\sigma_y + \sigma_z))$$

$$\epsilon_y = \frac{1}{E}(\sigma_y - \mu(\sigma_x + \sigma_z))$$

$$\epsilon_z = \frac{1}{E}(\sigma_z - \mu(\sigma_y + \sigma_x))$$

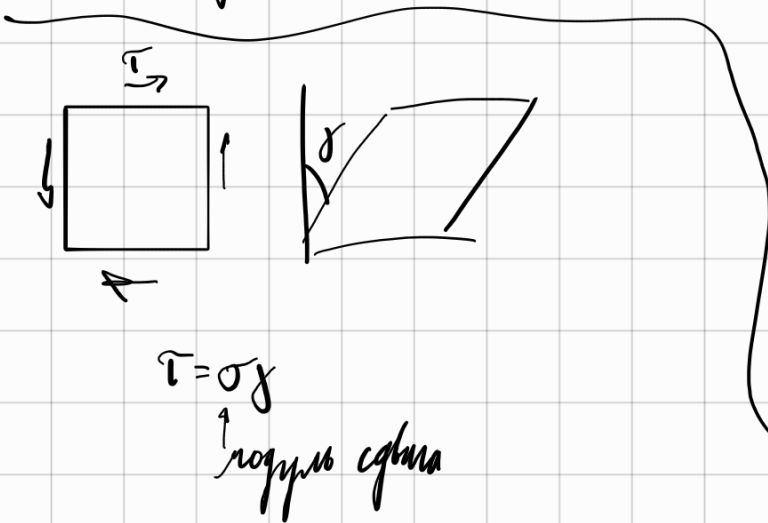


$$\sigma_x = \frac{4F}{\pi D^2}$$

$$\epsilon_x = \frac{4F}{E\pi D^2} - \frac{1}{E}2\sigma$$

$$\epsilon_y = \epsilon_z = 0 = \frac{1}{E}(\sigma - \mu(\frac{4F}{\pi D^2} + \sigma))$$

$$\sigma_y = \sigma_z = 0$$



$$\frac{\sigma}{E} = \frac{1}{E\pi D^2} + \frac{\sigma}{E}$$

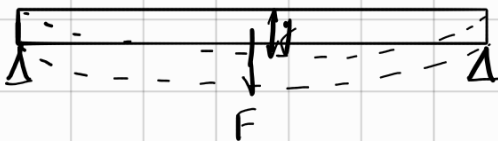
$$\sigma(\frac{1}{E} - \frac{1}{E}) = \frac{1}{E\pi D^2}$$

$$\sigma = \frac{1}{\pi D^2(1-\mu)}$$

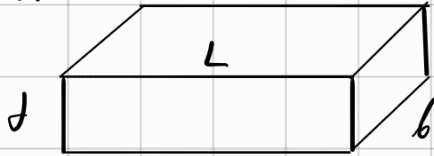
$$\epsilon_x = \frac{4F}{\pi D^2 E} - \frac{2}{E} \frac{1}{\pi D^2(1-\mu)} =$$

$$= \frac{4F}{\pi D^2 E} \frac{1-\mu-2\mu^2}{1-\mu} = \frac{1}{E}$$

полезно знать  
смайл



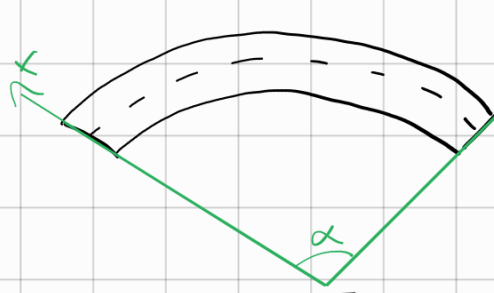
13.41



Канало:  $\alpha = 2\pi$   $R = \frac{L}{2\pi}$

$$\int W(x) = \frac{2\pi^2 E b}{L} x^2 dx$$

$$W = \int_{-\frac{d}{2}}^{\frac{d}{2}} \frac{2\pi^2 E b}{L} x^2 dx = \frac{2\pi^2 E b}{L} \frac{x^3}{3} \Big|_{-\frac{d}{2}}^{\frac{d}{2}} = \frac{\pi^2 E b d^3}{6L}$$



$$L = R\alpha$$

$$L(x) = (R+x)\alpha$$

$$\epsilon(x) = \frac{L(x) - L}{L} = \frac{x}{R}$$

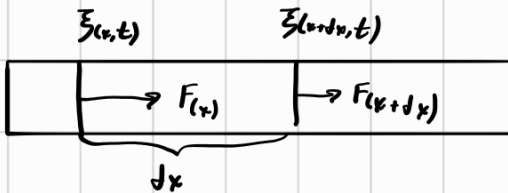
$$\sigma(x) = E \frac{x}{R}$$

Плотность энергии упругой деформации:

$$w(x) = \frac{\sigma(x)^2}{2E} = E \frac{x^2}{2R^2}$$

$$\int W(x) = w(x) \cdot L \cdot b \cdot dx = E L b \frac{x^2}{2R^2} dx$$

Волны



$$F(x+dx) \approx F(x) + \frac{\partial F}{\partial x} dx$$

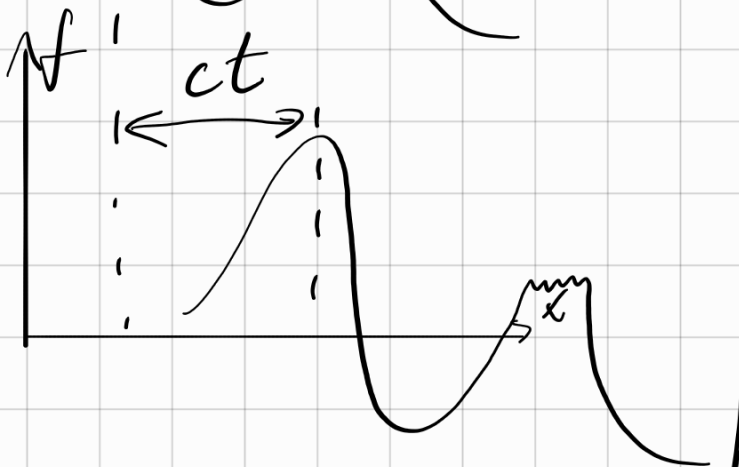
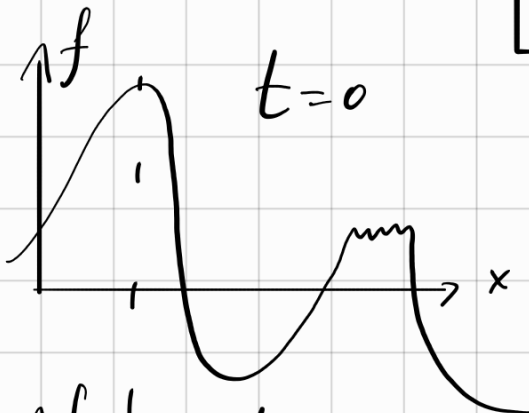
$$\int m \frac{\partial^2 \xi}{\partial t^2} = \frac{\partial F}{\partial x} dx$$

$$\int m = \rho S dx$$

$$\frac{F(x)}{S} = \sigma(x) = E \epsilon(x) = E \frac{\xi(x+dx, t) - \xi(x, t)}{dx} = E \frac{\partial \xi}{\partial x}$$

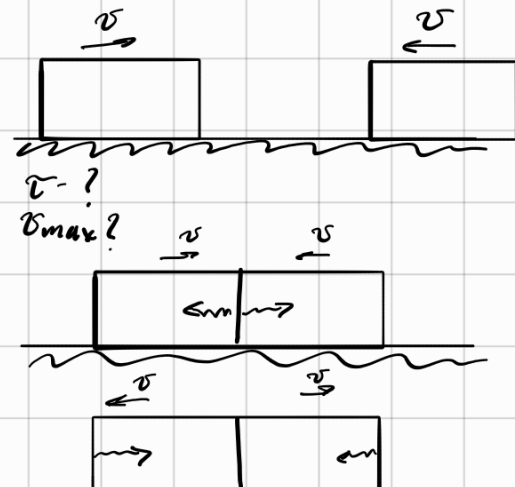
$$\rho S dx \frac{\partial^2 \xi}{\partial t^2} = S E \frac{\partial^2 \xi}{\partial x^2} dx$$

$$\boxed{\frac{\partial^2 \xi}{\partial t^2} = c^2 \frac{\partial^2 \xi}{\partial x^2}} \quad \text{волновое уравнение } c = \sqrt{\frac{E}{\rho}}$$



$$f(x - ct)$$

13.39



$$\tau = \frac{2l}{c} = 2l \sqrt{\frac{\rho}{E}} \approx 4 \cdot 10^{-5}$$

$$E \frac{\Delta L}{L_0} = T_{\text{spring}}$$

$$\Delta L = v_{\text{max}} \frac{T}{2}$$

$$\frac{v_{\text{max}}}{c} = \frac{T_{\text{spring}}}{E}$$

$$v_{\text{max}} \approx 5 \text{ m/s}$$